GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Data Collection Methods

Designanddemonstrationofdatacollectionmethodsandtools:experiment,questionnairesurvey, interviews and secondary data

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Research problem used in these activities: Comparative Search Efficiency of a Randomized Skip List, a DeterministicRed-BlackTreeandaSelf-AdjustingSplayTreeunderDifferentInsertionOrdersandAccess Patterns.

1. Aim

To design and demonstrate different **data collection methods and tools** (experiment, survey/questionnaire, interview, secondary data) related to the research problem. The **experiment** was actually carried out and its data are shown; the **questionnaire, interview guide and secondary-data plan** are fully designed instruments that are ready for use (no survey responses are presented in this report).

2. Overview of Data Collection Methods

Method / tool	Type of data	Whatitcollectsinthis study	Strengths	Limitations
Controlled experiment (measurement harness)	Primary, quantitative	Nodes inspected and timepersearchforeachstructure, order and workload	Objective, repeatable,controls extraneous factors	Resultsdependon theimplementation and machine
Questionnaire survey	Primary, mainly quantitative	Students'/developers'familiarity withandattitudes to randomized vs deterministic structures	Largesamplesat low cost; standardised answers	Self-reportbias; response rate; wording effects
Semi-structured interview	Primary, qualitative	Reasonsbehindchoiceof data structures in practice	Depth,explanations, new ideas	Time-consuming; small samples; interviewereffects
Observation / logs	Primary	Time taken by students to implementeachstructure(lab logs)	Recordsactual behaviour	Observereffect; effort to code
Secondarydata/ documents	Secondary	Published papers, library documentation,benchmark reports	Cheap, wide coverage,historical evidence	Notdesignedfor this question; quality varies

Measurement scales in this study

Scale	Property	Example in this study
Nominal	Categories without order	Datastructure(skiplist/red-black/splay);insertionorder; yes/no answers
Ordinal	Ordered categories	Likertresponses(1–5);yearofstudy;rankingofselection criteria
Interval	Equal intervals, no true zero	Notused(scoresofLikertsumsaretreatedasapproximately interval)
Ratio	Equal intervals and true zero	Nodesinspectedpersearch;μspersearch;yearsof programming experience

3. Method A – Controlled Experiment (carried out)

Design of the data collection

Element	Specification
Objective	Measuresearchcostofthreestructuresundercombinationsofinsertionorderandquery pattern.
Factors	Structure (3) × insertion order (2) × query pattern (2) = 12 conditions.
Replications	30 independent replications per condition (random seeds 1–30) →360 observations.
Fixed settings	n=5000keysdrawnfrom1–999,999;3000successful-searchqueriesperrun;skip-listp=0.5; same code base, machine and Python version.
Primary measure	Mean nodes inspected per search (counted inside the code: machine independent).
Secondary measure	Meanμspersearchmeasuredwith`time.perf_counter()`overthe3000queries(machine dependent).
Randomisation	Keys,querysequenceandskip-listlevelsaregeneratedfromseededrandomgenerators;the order of conditions is the same for every replication.
Recording	One CSV row per run, written automatically (no manual transcription) →file `RM_experiment_data.csv`.
Environment recorded	Python 3.12, Linux x86-64 (printed by the harness at the end of each run).

Instrument: measurement harness (excerpt)

The counting rules: skip list – every key comparison on a forward pointer; red-black tree – every node visited on the search path; splay tree – every node inspected while splaying. The complete code is delivered as `RM_measurement_harness.py`.

```
def run_queries(structure, s, queries):
    t0 = time.perf_counter()
    ifstructure=="SkipList":
        s.steps = 0
        forqinqueries: s.search(q)
        nodes=s.steps
    elif structure == "RedBlack":
        nodes = 0
        forqinqueries:
            nodes+=s.search_count(q)
    else:
        s.visits=0
        forqinqueries: s.search(q)
        nodes=s.visits
    dt=time.perf_counter()-t0
    returnnodes/len(queries),dt/len(queries)*1e6
```

```
defmake_data(rep,n,Q):
    rng=random.Random(1000+rep)
    keys = rng.sample(range(1, 10 ** 6), n)hot
    = rng.sample(keys, max(1, n // 100)) uni =
    [rng.choice(keys) for _ in range(Q)]
    skew = [rng.choice(hot) if rng.random() < 0.9 else rng.choice(keys) for _ in range(Q)]
    return keys, {"uniform": uni, "skewed": skew}
```

Data collected (extract of the raw data file)

replication	structure	order	workload	n	Q	nodes/search	μs/search
1	SkipList	random	uniform	5000	3000	23.81	1.77
1	SkipList	random	skewed	5000	3000	24.88	1.85
1	RedBlack	random	uniform	5000	3000	11.61	0.77
1	RedBlack	random	skewed	5000	3000	11.66	0.64
1	Splay	random	uniform	5000	3000	21.38	2.63
1	Splay	random	skewed	5000	3000	10.35	1.34
1	SkipList	sorted	uniform	5000	3000	23.56	1.96
1	SkipList	sorted	skewed	5000	3000	24.47	1.75
1	RedBlack	sorted	uniform	5000	3000	11.94	0.72
1	RedBlack	sorted	skewed	5000	3000	12.67	0.69
1	Splay	sorted	uniform	5000	3000	23.55	2.56
1	Splay	sorted	skewed	5000	3000	12.87	1.47

First12of360rows.Variables:replication,structure,insertion_order,workload,n_keys,n_queries, mean_nodes_per_search, us_per_search.

Data quality checks

- **Completeness:** 360 rows = 3 structures × 2 orders × 2 workloads × 30 replications; no missing values (0 found).
- **Range check:** nodes per search between 10.14 and 29.55 (plausible for n=5000 where $\log_2 n \approx 12.3$).
- **Correctness check:** the implementations were verified against a reference set with randomized insert/delete/search tests before data collection.
- **Reproducibility check:** running replication 1 forth e skip list twice gave identical node counts (23.8083 and 23.8083), while wall-clock times differed (2.946 and 2.673 μs), showing that the node-count measure is deterministic given the seed and the timing measure is subject to noise.

Measurement uncertainty analysis

Uncertaintyofthemeanof30replicationswasestimatedfromthereplication-to-replicationvariation(Type A evaluation): standard error = SD / √30, and a 95% confidence half-width = $t_{0.975,29}$ × SE (t = 2.045). Uniform queries, random insertion:

Structure	Mean (nodes)	SD	Standard error	95% half-width	Relative uncertainty
SkipList	24.315	1.689	0.308	± 0.631	2.59%
RedBlack	11.622	0.036	0.007	± 0.014	0.12%
Splay	21.255	0.086	0.016	± 0.032	0.15%

The relative uncertainty of the node-count measure is below about 3% for every structure. The timing measure is much noisier: for the red-black tree the μs/search values ranged from 0.61 to 1.35 across replications (skip list: 1.62–3.24), because of operating-system scheduling, caches and the Python interpreter. For this reason node counts were chosen as the primary measure. Systematic uncertainty (e.g., counting rules that differ slightly between structures) cannot be removed by repetition and is discussed as a threat to validity in the Research Report.

4. Method B – Questionnaire Survey (instrument designed)

Survey plan

Element	Specification
Title	Data Structure Selection Practices and Perceptions Survey
Objective	Todescribehowstudents/developersratetheirknowledgeofbalanced,self-adjustingand randomized structures and what criteria they use to choose among them (supports the interpretation of the experiment).
Target population	Postgraduate computer-engineering students and junior developers (assumed N = 600).
Sampling plan	Stratifiedrandomsamplebyyearofstudy;n=235fromCochran'sformulawith finite-population correction (ALA 3).
Mode and tool	Onlineform(GoogleForms/MicrosoftForms);approx.8minutes;responsesexportedto CSV.
Ethics	Voluntary,anonymous;consentstatementonfirstpage;nopersonalidentifiers;dataused only for the study.

Questionnaire (items)

Consent: “I have understood the purpose of this survey, my participation is voluntary and my answers will remain anonymous.” [ ] I agree [ ] I do not agree

ID	Question / statement	Response format	Scale
	Section A – Respondent profile		
A1	Current year of study	Singlechoice:M.TechYear1/ Year 2 / Working professional	Nominal
A2	Years of programming experience	Number	Ratio
A3	Haveyouimplementedabalancedsearchtree yourself? A skip list?	Yes / No for each	Nominal
	SectionB–Familiarity(1=notfamiliar,5=very familiar)		
B1–B4	Howfamiliarareyouwith:(B1)red-blacktrees,(B2) skip lists, (B3) splay trees, (B4) hash tables?	5-point scale (one per item)	Ordinal
	Section C – Choice of data structure		
C1	Rank these factors for choosing an ordered data structure (1 = most important): worst-case guarantee, simpleimplementation,speedinpractice,memoryuse, adapts to access pattern	Ranking	Ordinal
C2	Whichstructuredidyouuseinyourlastprojectthat needed ordered data?	Singlechoice:librarytreemap/ skip list / sorted array / other	Nominal
	SectionD–Perceptions(1=stronglydisagree…5= strongly agree)		

ID	Question / statement	Response format	Scale
D1	Randomizeddatastructuresarelessreliablethan deterministic ones. (reverse-coded)	5-point Likert	Ordinal
D2	Randomizationisagoodwaytoavoidworst-case inputs.	5-point Likert	Ordinal
D3	Iwoulduseaskiplistifmyprogramminglibrary provided one.	5-point Likert	Ordinal
D4	Performanceofadatastructuredependsmainlyon how the data is accessed.	5-point Likert	Ordinal
D5	Iconsidertheaccesspatternbeforechoosingadata structure.	5-point Likert	Ordinal
D6	Itrustperformanceclaimsonlywhenexperimentsare repeated many times.	5-point Likert	Ordinal
	Section E – Open-ended		
E1	Describe a situation in which the choice of data structurechangedtheperformanceofyourprogram.	Free text	Qualitative

Constructs, validity and reliability

Construct	Items	Planned analysis
Attitudetowardsrandomized structures	D1(reverse-coded:6−score),D2, D3	Mean score; Cronbach's alpha
Workload awareness	D4, D5	Meanscore;correlationbetweenD4 and D5
Familiarity	B1–B4	Meanperstructure;Friedmantest between B1–B3

- **Content validity:** two faculty members and one industry developer review each item for relevance and clarity.
- **Face validity / pilot:** test on 10–15 students not in the sample; note unclear wording, completion time, and drop-outs; revise.
- **Reliability:**internalconsistencybyCronbach'salpha,α**=k/(k−1)·(1−**$\Sigma\sigma_i^2/\sigma_T^2$**)**wherekisthe numberofitems,σ_i^2thevarianceofitemiandσ_T^2thevarianceofthetotalscore.Avalueofatleast 0.70 is accepted; otherwise items with low item-total correlation are revised or removed.
- **Biascontrol:**neutralwording,oneideaperstatement,areverse-codeditemtodetectcareless answering, randomised order of the Section D items.

```
# Cronbach's alpha for an items matrix X (rows = respondents, columns = items)
import numpy as np
defcronbach_alpha(X):
    X = np.asarray(X, dtype=float)
    k = X.shape[1]
    item_var = X.var(axis=0, ddof=1).sum()
    total_var = X.sum(axis=1).var(ddof=1)
    returnk/(k-1)*(1-item_var/total_var)
```

5. Method C – Semi-structured Interview (guide designed)

Element	Specification
Participants	6–8purposivelyselectedexperienceddevelopers/faculty(judgementsampling),untilnonew themes appear (saturation).
Durationand recording	20–30 minutes; audio recorded with permission; transcribed.
Questions	1. Which ordered data structures have you used in real projects, and why? 2. Whatmattersmostwhenyouchoosebetweenaguaranteedstructureandarandomized one? 3. Have you seen performance problems caused by a poor choice or by the workload? 4. How do you measure and compare performance in your work? 5. Would you trust a randomized structure in a critical system? Why / why not? 6. Anything else you want to add?
Probes	'Can you give an example?', 'What happened next?', 'Why do you think so?'
Analysis plan	Transcribe →read →code segments →group codes into themes (thematic analysis) → compare themes with the experimental findings.

6. Method D – Secondary Data and Documents (plan)

Source	Data to extract	Quality check
Peer-reviewedpapers(ALA1 list)	Theoreticalbounds,reportedexperiments, evaluation methods	Peerreview;checkagainst original publisher record
Official library documentation	Whichstructureisusedbystandardlibraries(for example, Java `TreeMap`is a red-black tree and `ConcurrentSkipListMap`is a skip list)	Usetheofficialdocumentation of the specific version
Open-sourcerepositories/ system documentation	Real systems that use skip lists or trees (for example,Redisdocumentsskiplistsforitssorted sets)	Citerepositoryversion/date; avoid unofficial blogs

7. Data Management and Integration

- **Triangulation:**experimentalmeasurements(whathappens),survey(whatpeoplebelieve/do)and interviews (why) are compared to strengthen conclusions.
- **Fileformatsandnaming:**rawdatastayunchangedinCSV(`RM_experiment_data.csv`); processed files and code are stored separately with version/date.
- **Backup and sharing:** two copies (local and cloud); only anonymised survey data are shared.
- **Documentation:** a data dictionary (variable names, units, scale) accompanies every data file.

Data dictionary of the collected experiment data

Variable	Description	Type / unit
replication	Replication number (random seed group)	Integer 1–30
structure	SkipList, RedBlack or Splay	Nominal
insertion_order	random or sorted order of inserting the keys	Nominal
workload	uniform or skewed query pattern	Nominal
n_keys, n_queries	Number of stored keys and of queries per run	Count
mean_nodes_per_search	Average nodes inspected per search	Ratio (count)
us_per_search	Average microseconds per search (Python wall-clock)	Ratio (μs)

8. Conclusion

Four data collection methods were designed for the research problem. The controlled experiment was executed and produced a complete, quality-checked data set of 360 observations with quantified measurement uncertainty. The questionnaire, interview guide and secondary-data plan are complete instruments, with sampling, validity, reliability and ethics provisions, ready to be administered. Using several methods together allows the quantitative results to be interpreted with practitioner views.

References

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